Risk and Return Part 2

Chapter 7

Introduction to Financial Management (Pope)

Technical Note on Variance

Historical/real data is often considered uniform in distribution, implying each observation has the same probability (1/n), so,

$$population\ var(X) = \frac{1}{N} * \sum_{i=1}^{N} (X_i - \bar{X})^2 = \sigma_{pop}^2$$

The above formula is for a population variance (you have all the observations) but most of the time we only have a sample, so,

$$sample\ var(X) = \frac{1}{N-1} * \sum_{i=1}^{N} (X_i - \bar{X})^2 = \sigma_{sample}^2$$

Verizon (VZ)

The table present the annual returns of Verizon Communications over the a 5 year period.

Year	Annual Return	Deviations from Average Return	Squared Deviation
2017	3.98%	-0.00302	0.00001
2018	11.26%	0.06980	0.00487
2019	13.84%	0.09558	0.00914
2020	-0.13%	-0.04409	0.00194
2021	-7.55%	-0.11827	0.01399
Total	21.39%	0	0.02995

$$\mathrm{E}(r_{VZ}) = \sum_{i=1}^{5} p_i X_i = \frac{21.39\%}{5} = 4.28\%$$

$$var(r_{VZ}) = \frac{1}{N-1}\sum_{i=1}^{N}(X_i - \bar{X})^2 = \frac{0.02295}{5-1} = 0.0075$$

$$sd(r_{VZ}) = \sqrt{0.0075} = 0.0865 = 8.65\%$$

Ticker	Company	SD (%)	Ticker	Company	SD (%)
X	U.S. Steel	76.4	UNP	Union Pacific	21.4
PCG	Pacific Gas & Electric	55.4	DIS	Disney	21.2
MRO	Marathon Oil	48.9	GOOG	Alphabet	20.1
NEM	Newmont Mining	36.7	SBUX	Starbucks	18.7
AMZN	Amazon	28.3	WMT	Walmart	17.7
BA	Boeing	27.4	XOM	ExxonMobil	19.4
CPB	Campbell Soup	22.6	PFE	Pfizer	16.6
IBM	I B M	22.5	MCD	McDonald's	14.4
F	Ford	22.1	KO	Coca-Cola	12.6
INTC	Intel	22.1	**S&P 500**		**12.0**

Diversification

Diversification is a strategy designed to reduce risk by spreading the portfolio across many investments.

- By dividing up investments across **many relatively low-correlated assets,** it is possible to considerably reduce one's exposure to risk
- Markowitz (1959), "Portfolio Selection", *Journal of Finance*

$$r_{portfolio} = \sum_{i=1}^{N} w_i r_i$$

Example 1:

You are considering an investment into two companies and current market conditions predict a 40% chance of an economic boom and a 60% chance of a recession. Zig Inc is predicted to return 25% in a boom state and 3% in a recession. Zag Co. is estimated to have an 8% return in a boom and a 23% return in a recession. What are the risk-return profiles of each company and what if you build an equally weighted portfolio of the two?

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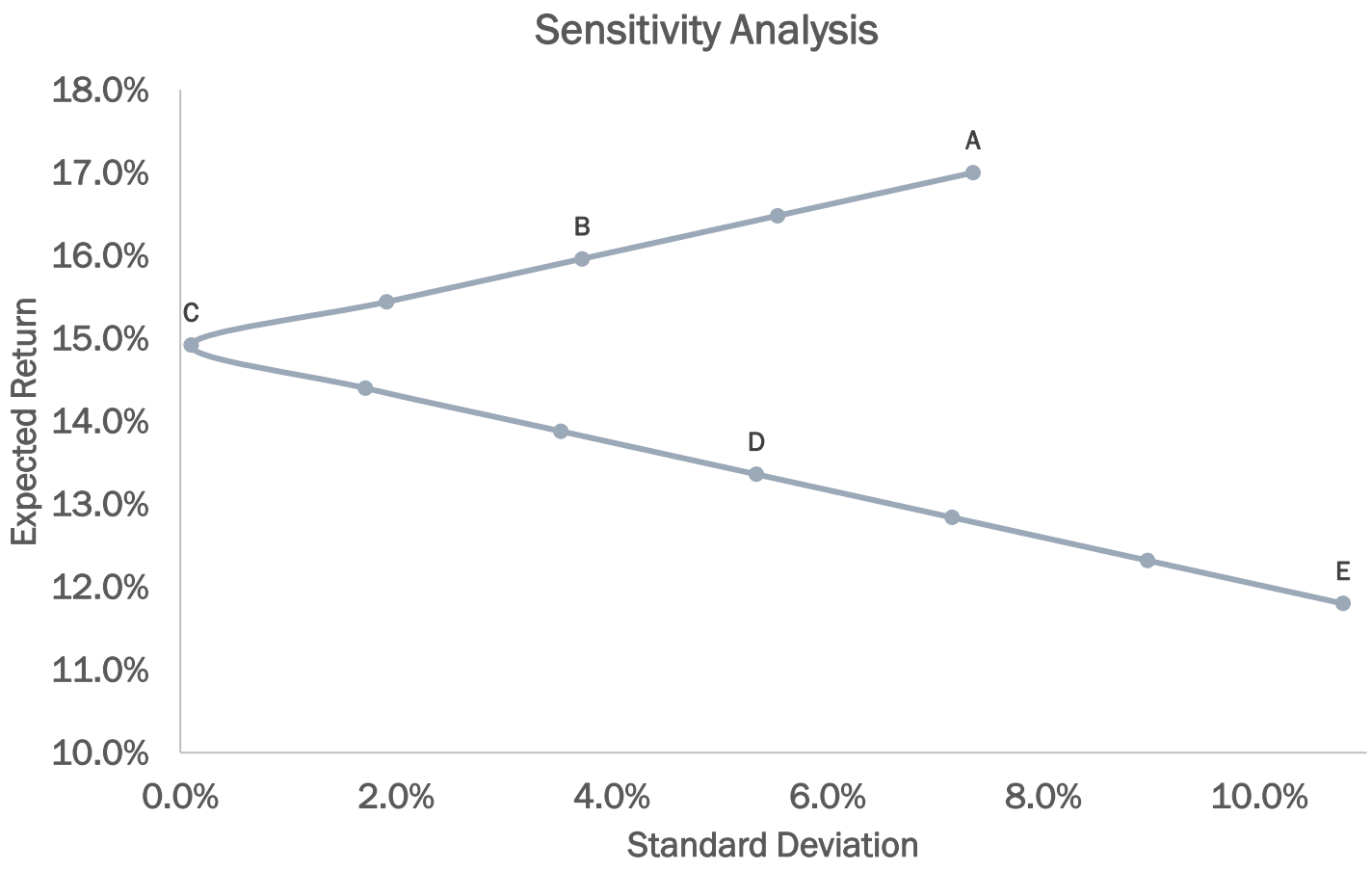
Scenario	Probability	Zig Inc	Zag Co	50-50 Portfolio
Boom	40%	25%	8%	16.5%
Recession	60%	3%	23%	13%
Expected Return		11.8%	17%	14.4%
Standard Deviation		10.78%	7.35%	1.71%

Example 1: Sensitivity Analysis

Moving from points A to E is a change in the weights of each asset in the portfolio.

A = 100% Zag, 0% Zig

E = 0% Zag, 100% Zig



Diversification Benefit

Idiosyncratic/Diversifiable risk:

- Related to asset (company) specific events/uncertainty
- Also called firm-specific risk

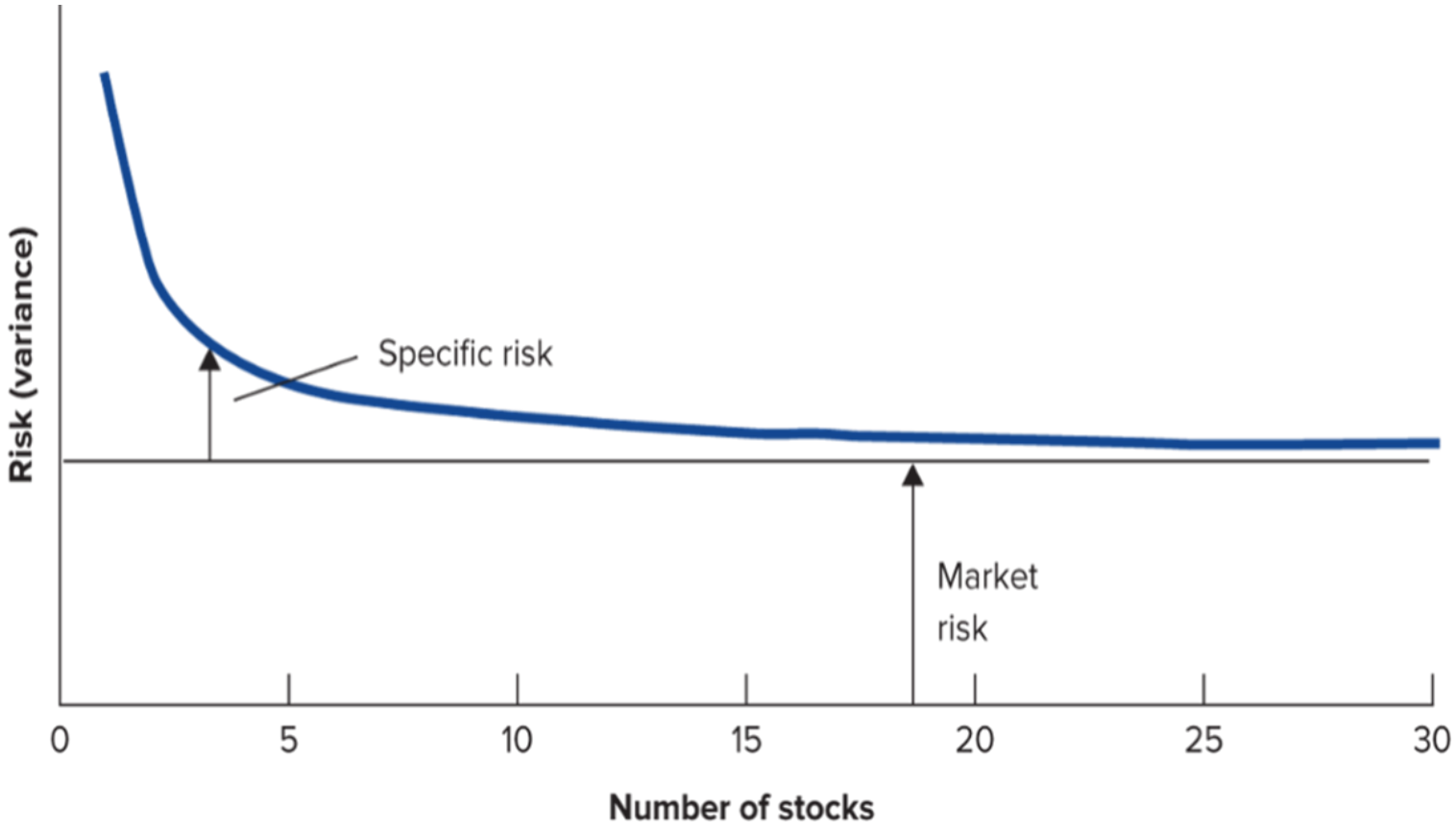
Systematic/Non-diversifiable risk:

- Economy-wide sources of risk that affect the overall stock market
- Also called market risk

Well-diversified portfolio: one whose specific risk has been eliminated

The "Market Portfolio"

Diversification benefit leads an investor to build the most well diversified portfolio possible, i.e., the market portfolio

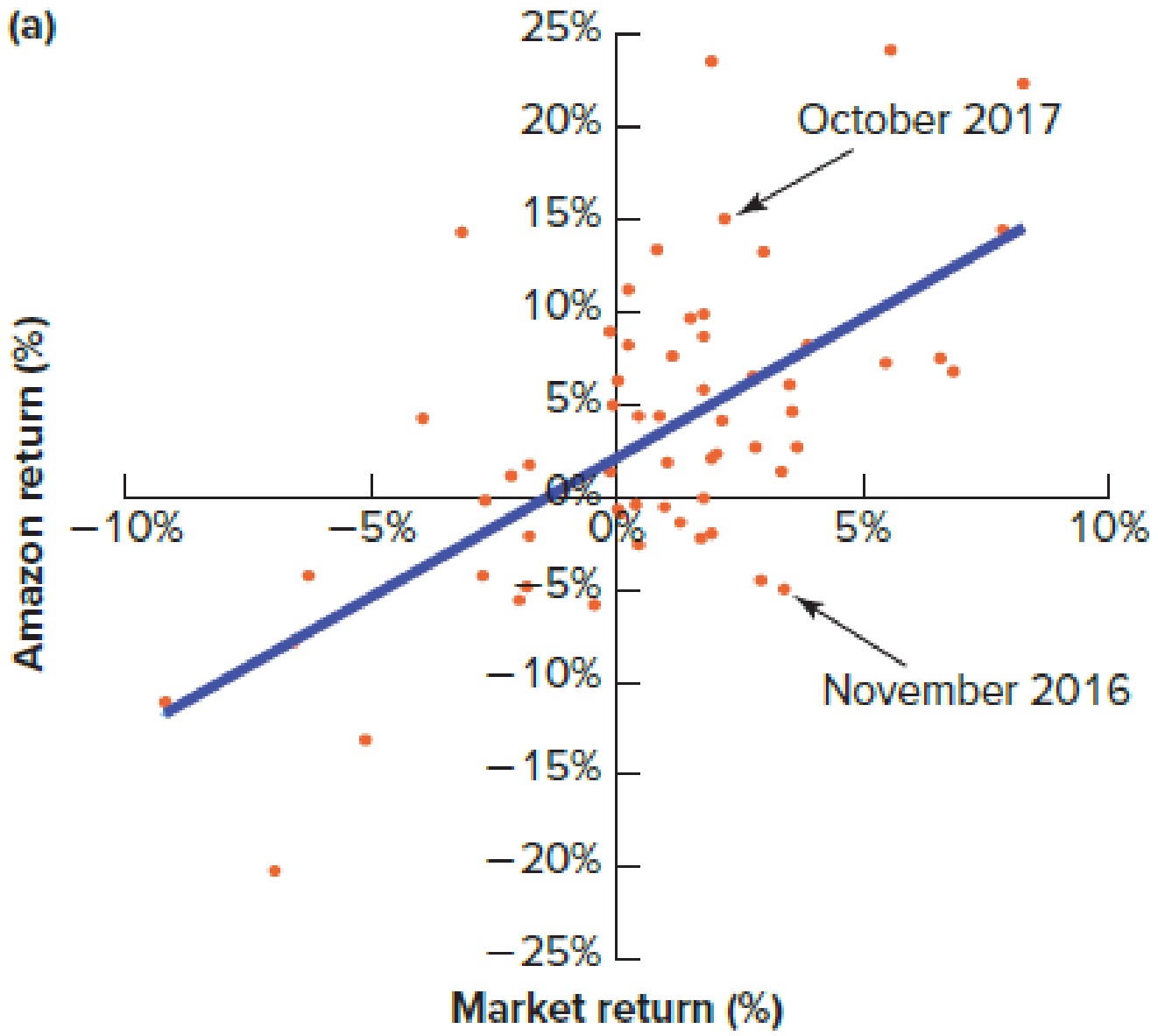


Beta

Beta measures the sensitivity of a stock's return to the return on the market portfolio

- It is a measure of risk for a "well-diversified portfolio"
- Typically calculated by a linear regression of stock returns on the overall market return (like the S&P 500)
- More roughly,

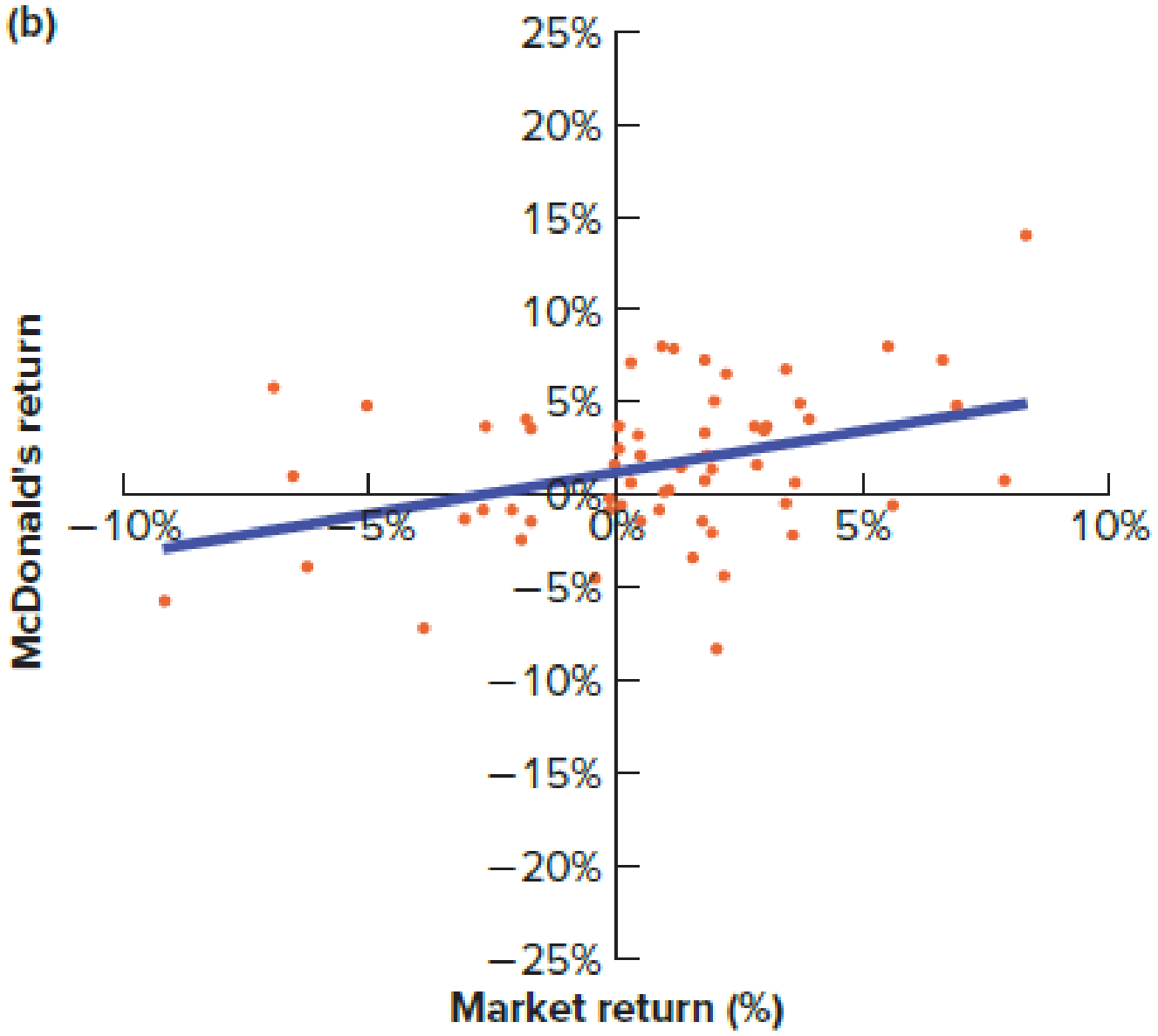
$$\beta_i = \frac{\Delta r_i}{\Delta r_{market}}$$



Amazon.com Inc

Returns on AMZN are plotted against returns from the S&P 500 from January 2015 through December 2019.

AMZN Beta = 1.51



McDonald's Corporation

Returns on MCD are plotted against returns from the S&P 500 from January 2015 through December 2019.

MCD Beta = 0.45

Betas in Practice

The tables present beta estimates for various components of the S&P 500

Ticker	Company	Beta
X	U.S. Steel	3.03
MRO	Marathon Oil	2.35
AMZN	Amazon	1.51
BM	IBM	1.33
BA	Boeing	1.19
F	Ford	1.09
UNP	Union Pacific	1.07
GOOG	Google/Alphabet	1.01
DIS	Disney	1.00
XOM	ExxonMobil	1.00

Ticker	Company	Beta
INTC	Intel	0.91
PFE	Pfizer	0.65
PCG	Pacific Gas & Electric	0.55
SBUX	Starbucks	0.51
MCD	McDonald's	0.45
CPB	Campbell Soup	0.43
KO	Coca-Cola	0.43
WMT	Walmart	0.35
NEM	Newmont Mining	−0.01
Average		0.99

Portfolio Beta

We can measure the risk of a well-diversified portfolio by measure the beta of the overall portfolio

- Aids in determining performance (return) relative to risk of a portfolio
- See how each asset contributes to the overall risk of the portfolio

$$\beta_{portfolio} = \sum_{i=1}^{N} w_i \beta_i$$

Capital Asset Pricing Model

The expected rate of return on an investment depends upon two things:

- Basic compensation for waiting (risk-free return)
- An investment-specific risk premium
- CAPM models the "well diversified" investors trade-off between risk and return (the ***required return*** based upon portfolio risk)

$$E(r_i) = r_f + \beta(E(r_m) - r_f)$$

Example 2:

Let's say that you are looking at investing in 2 stocks A and B. A has a beta of 1.3 and is predicted to have a return of 15%, B has a beta of 0.9 and is predicted to earn 9%. If the risk-free rate is currently 4% and the expected return on the market is 11%, determine whether these stocks are worth investing in.

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*Stock A's expected/required return = 4% + (11%-4%)*1.3 = 13.1%*

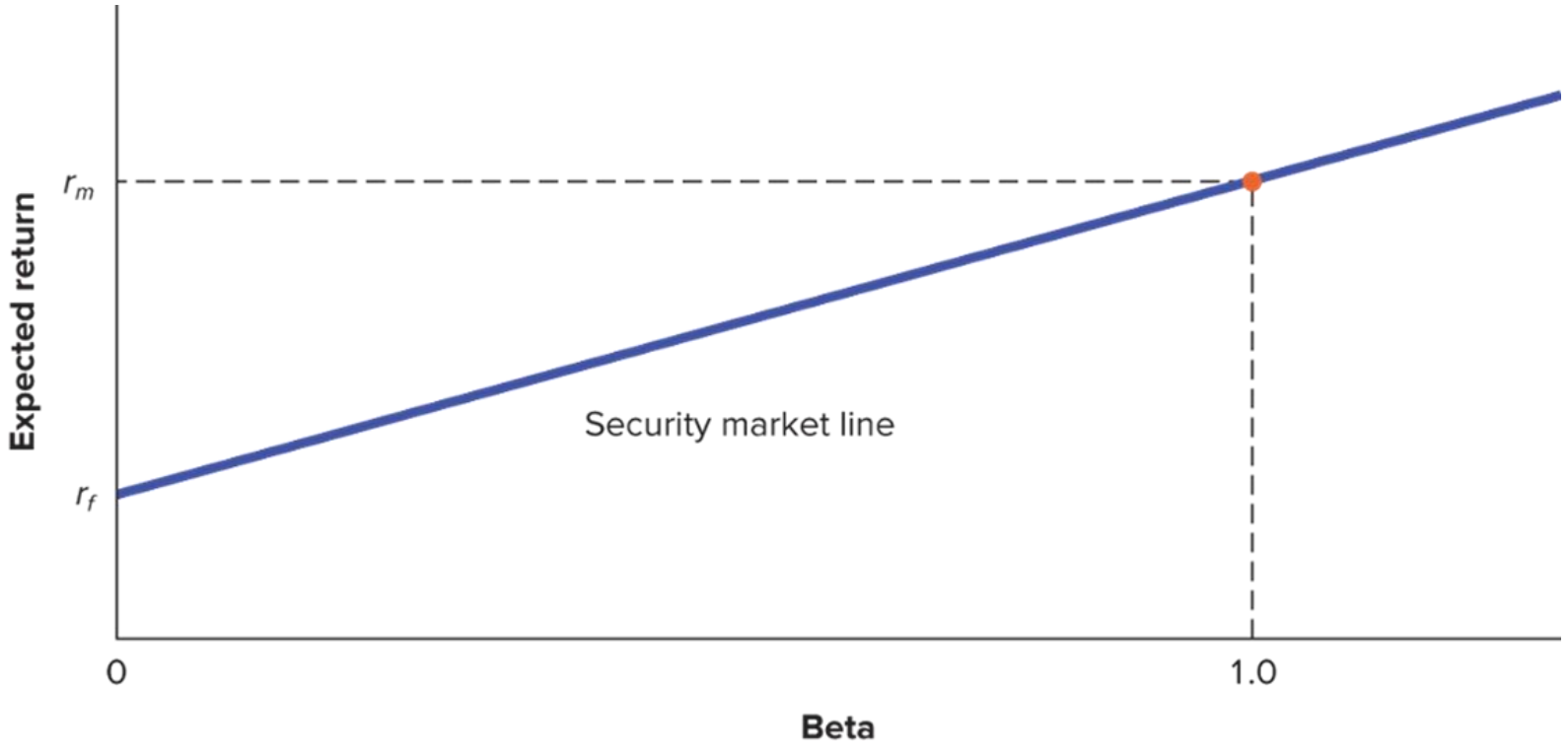
*Stock B's expected/required return = 4% + (11%-4%)*0.9 = 10.3%*

Stock A would, since 15%>13.1% would be considered undervalued (it offers a higher return that required), while stock B would be considered overvalued (9%<10.3%) since it offers a lower return than required.

Security Market Line

A graphical representation of the CAPM with a given risk-free rate and expected market return.

Used to evaluate the relative risk to return tradeoff of an equity.



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